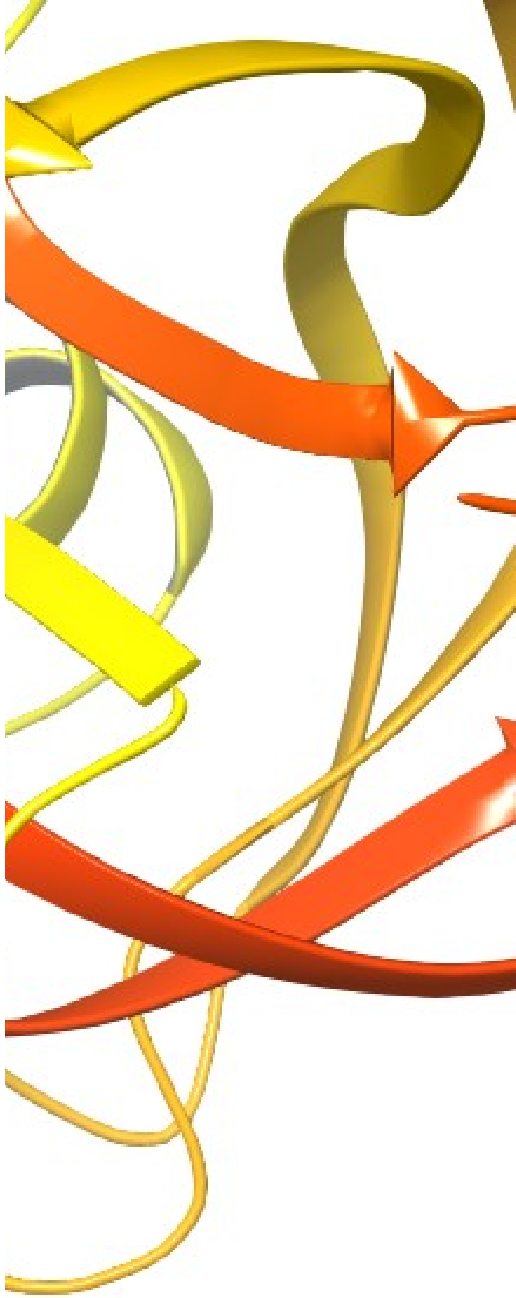
Glide Docking

Stephen Nnemolisa, Shadrach Eze



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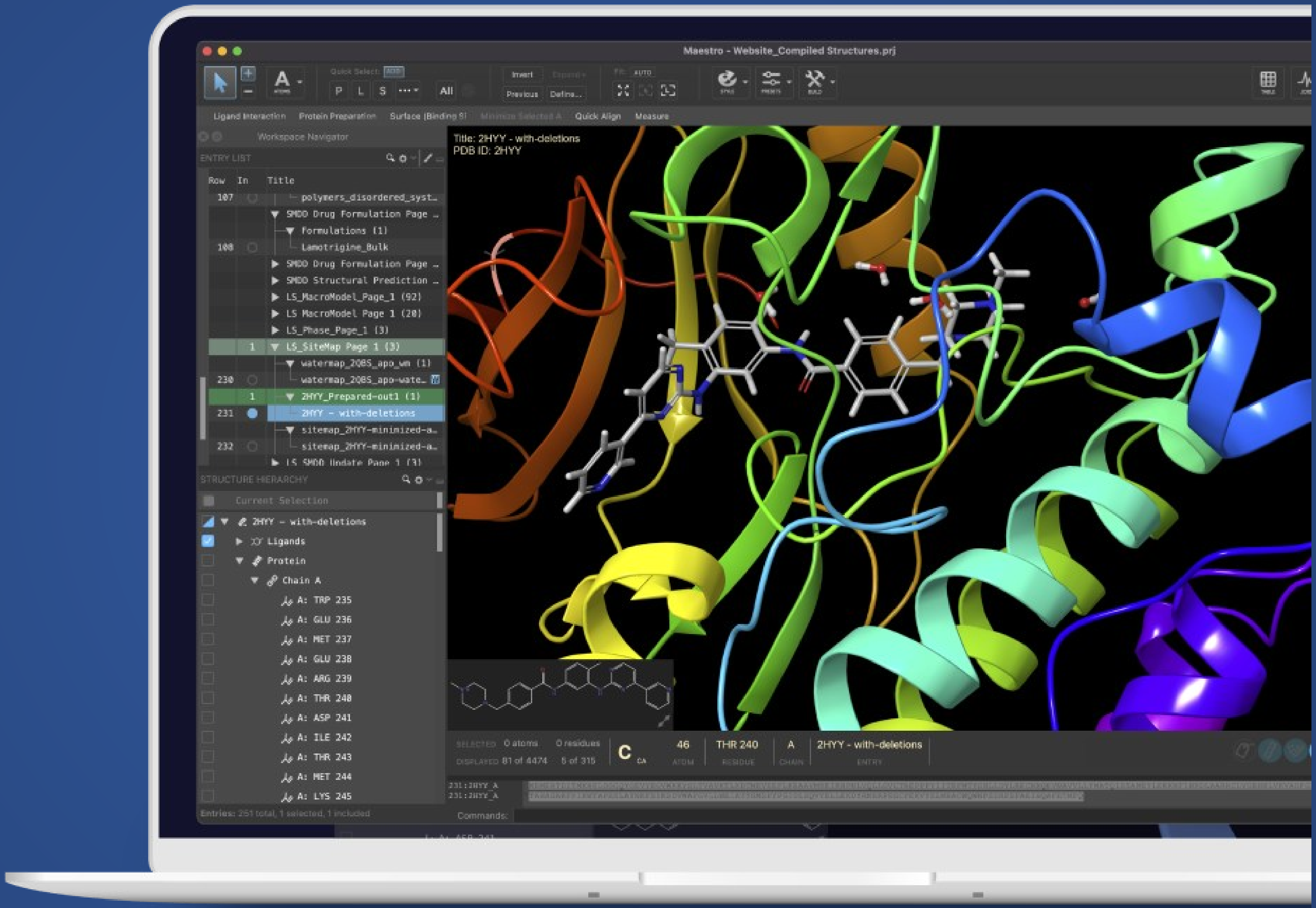
Maestro



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Workflow

- Import Structures
- Lig-Prep
- Protein Preparation
- Grid Generation
- Glide Docking
- Pose Visualization

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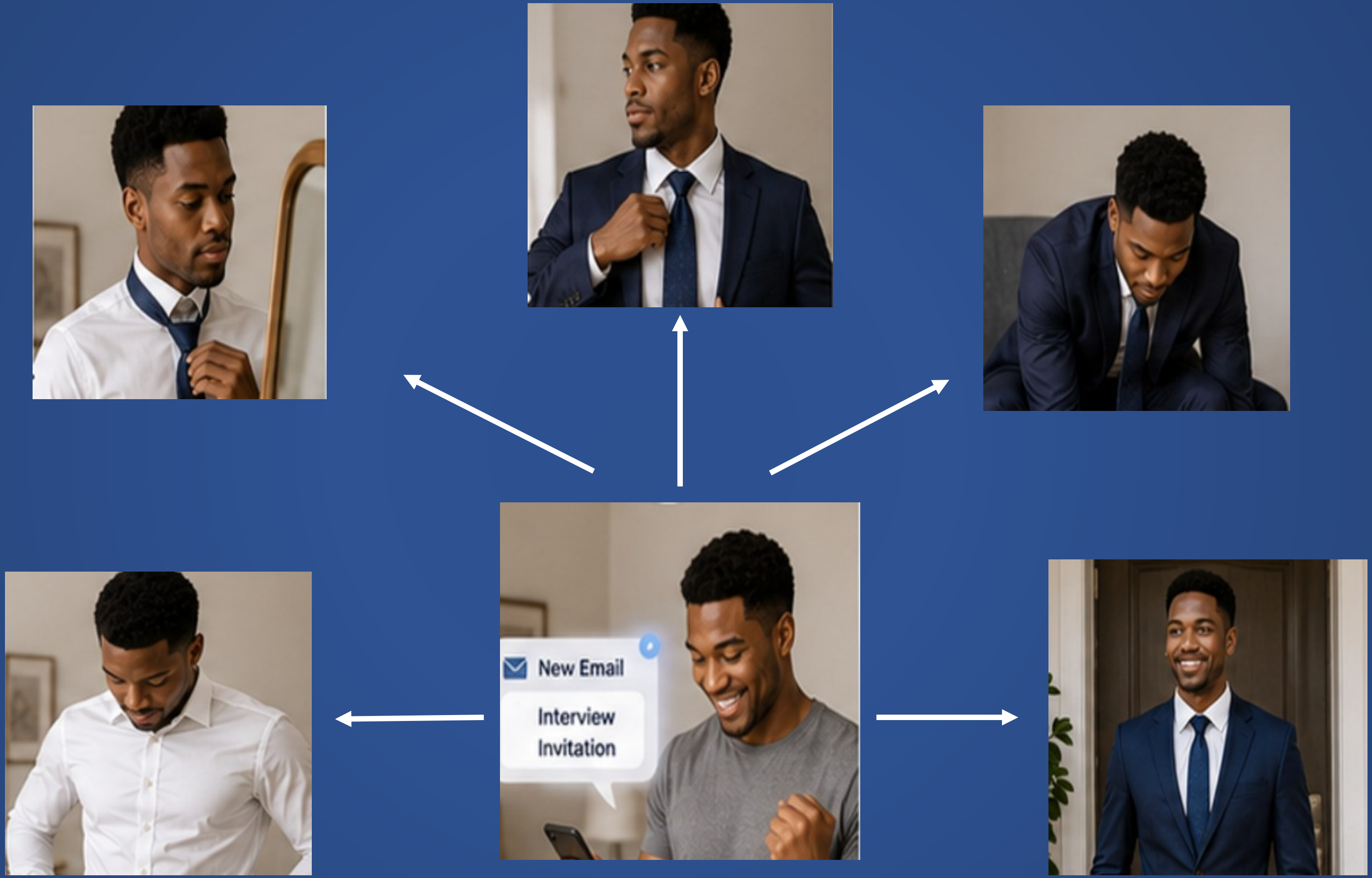
What is the purpose of Lig-Prep?



Ligand preparation is ideally performed either before initiating virtual screening or after library design

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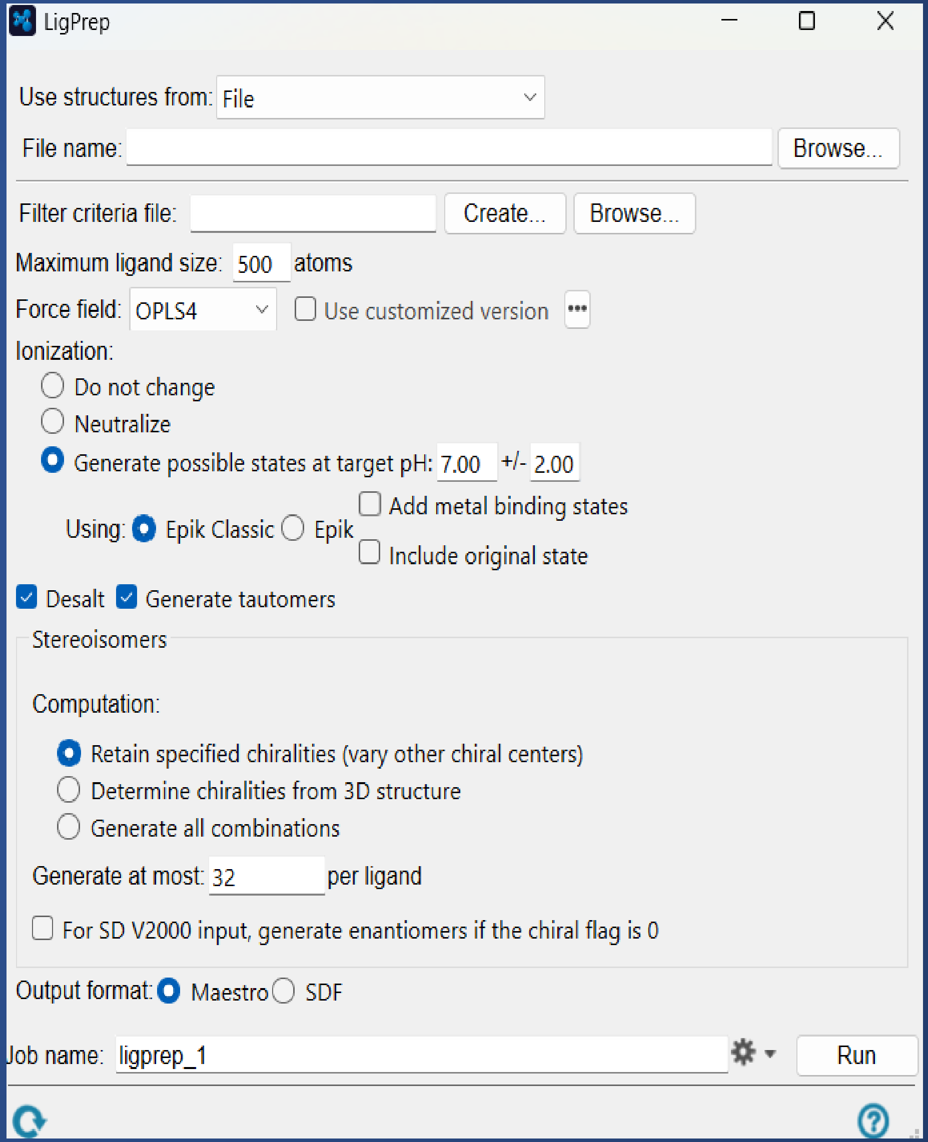
What is the purpose of LigPrep?

- **Lig-Prep** is a computational software tool developed by Schrödinger used in drug discovery and molecular modeling.
- It converts and optimizes 1D, 2D, or 3D chemical structures into high-quality, energy-minimized 3D all-atom structures ready for virtual screening and molecular docking

- **Lig-Prep** is highly valued and considered an industry-standard tool in drug discovery:
  - Seamless workflow integration
  - High-Quality 3D Geometries



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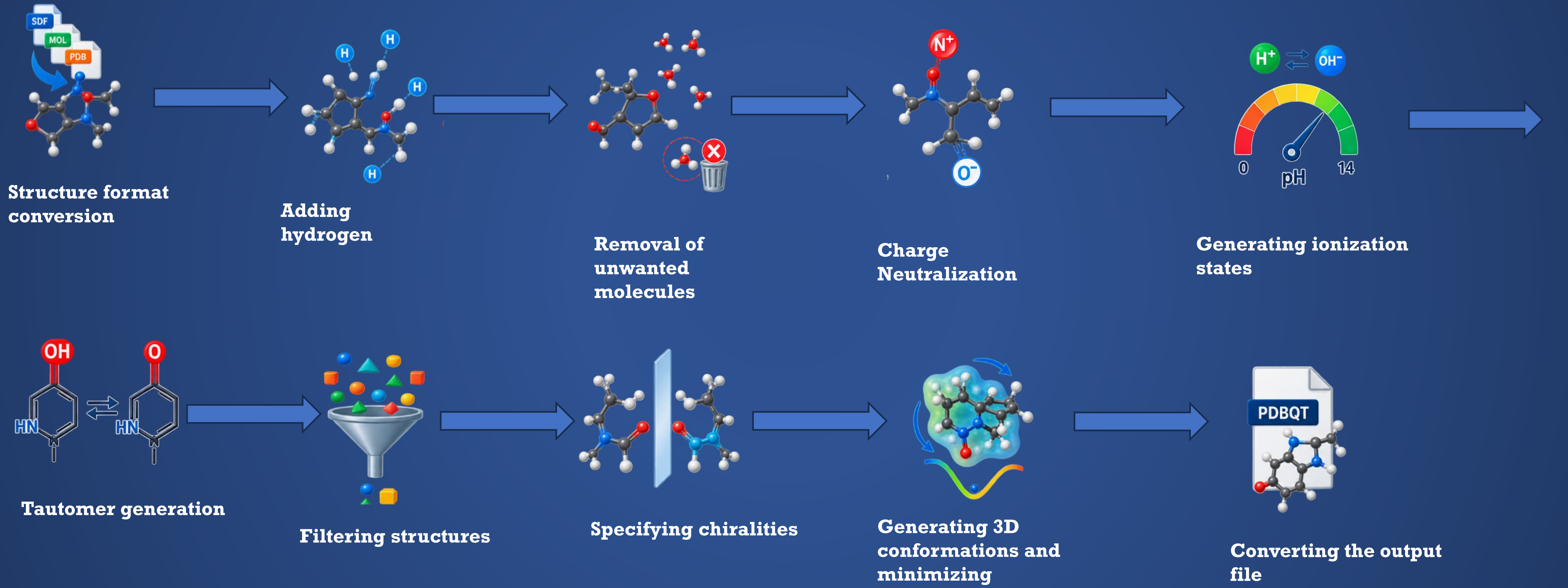
How does it work?



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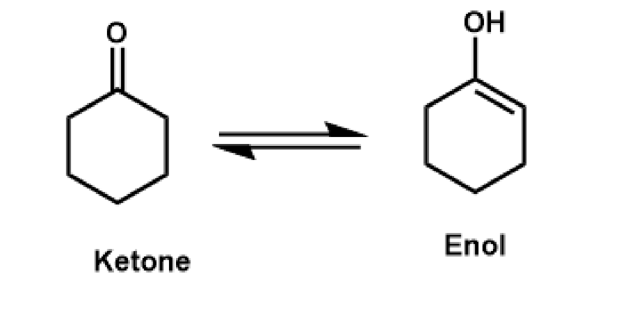
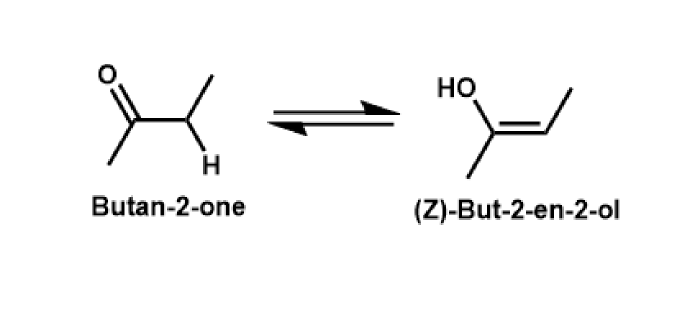
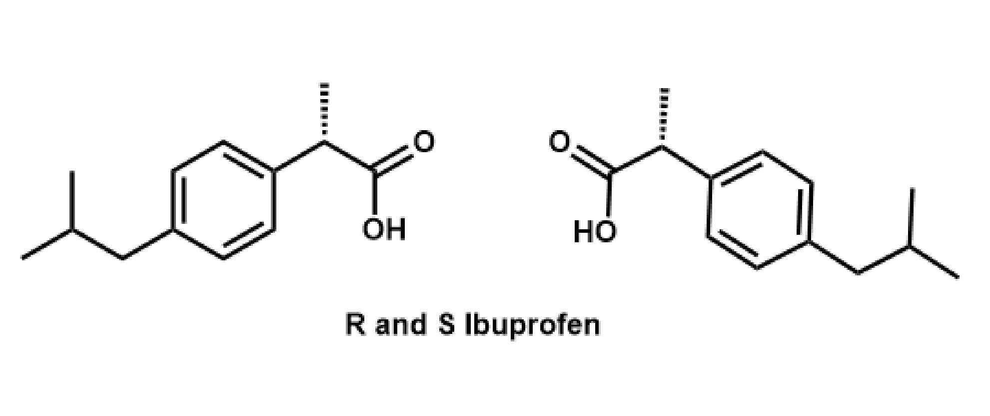
Important terms to keep in mind?

- **Tautomer:**
  - They are compounds that readily interconvert through either the movement of a hydrogen atom (proton) or shifts in bonding





- **Stereoisomers:**
  - Molecules that have the same chemical formula and atom connections, but a different shape in three-dimensional space.



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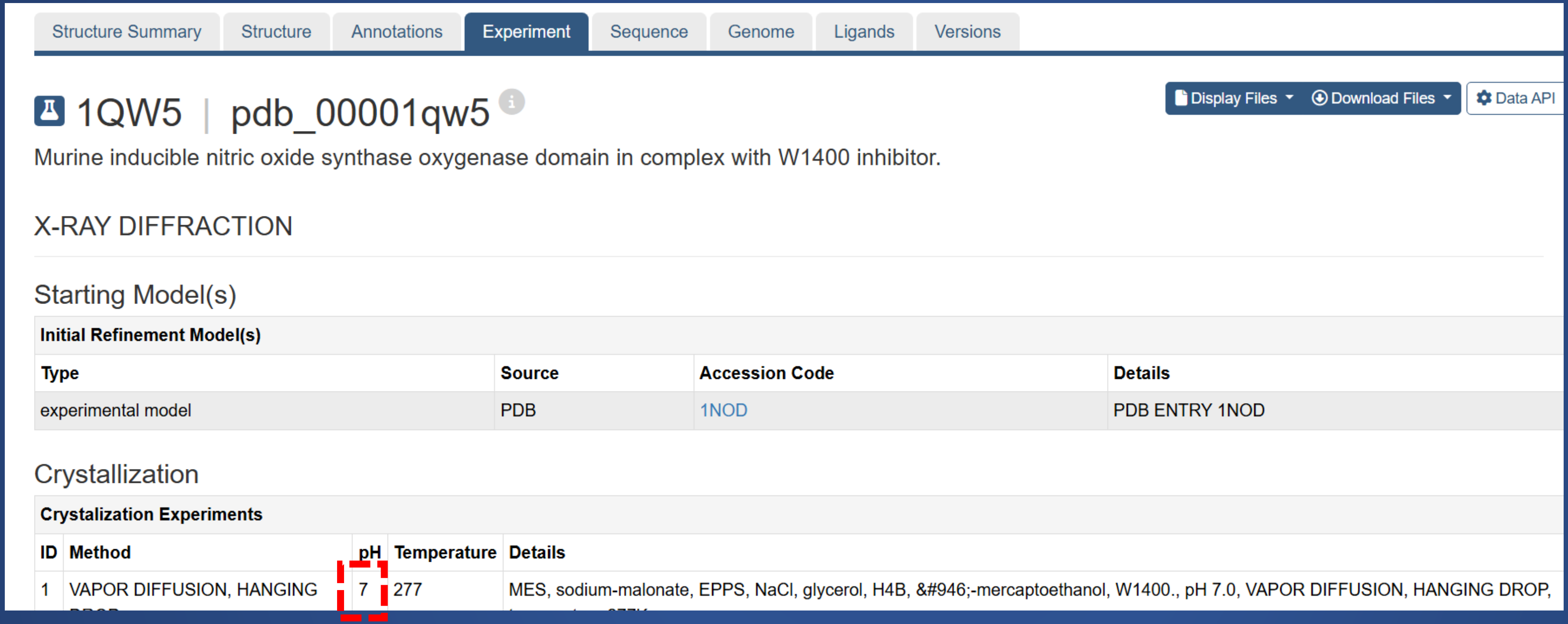
Some key Lig-Prep settings



Structure Summary | Structure | Annotations | Experiment | Sequence | Genome | Ligands | Versions

Display Files | Download Files | Data API

1QW5 | pdb_00001qw5

Murine inducible nitric oxide synthase oxygenase domain in complex with W1400 inhibitor.

X-RAY DIFFRACTION

Starting Model(s)

Initial Refinement Model(s)

Type	Source	Accession Code	Details
experimental model	PDB	1NOD	PDB ENTRY 1NOD

Crystallization

Crystalization Experiments

ID	Method	pH	Temperature	Details
1	VAPOR DIFFUSION, HANGING	7	277	MES, sodium-malonate, EPPS, NaCl, glycerol, H4B, β-mercaptoethanol, W1400., pH 7.0, VAPOR DIFFUSION, HANGING DROP,

- **Setting pH:** The pH range can be tailored to suit project requirements. **Lig-Prep** defaults to a neutral pH
- **Metal binding state:** This option can be enabled if your ligand involves known metal interaction.

Desalting, tautomer generation, and retention of the appropriate ionization state can be used to refine the molecular structure further

- Stereochemistry is another important consideration when setting a Lig-Prep job

Stereochemistry	When to use
Retain Specified chiralities	This could be used when your input is a 2D structure or a SMILES string, where some chiral centers have specified chirality (that you would like to retain) and others are not specified.
Determine chiralities from 3D structures	This could be used when your input is a 3D structure and are certain about the chirality of any chiral center
Generate all combinations	This could be used when the chirality was specified, but you are uncertain of the chirality of the chiral center.

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